

Aritrabha Majumdar

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Currently, I am a postgraduate student at Indian Statistical Institute, Kolkata. I am fascinated by the elegance and beauty of Probability and its application in numerous fields. My topics of interest include Probability Theory, Random Walk, Dynamical Systems, Random Polymers, Fractal Percolation and Random Tessellation.

Education

B.Math	Indian Statistical Institute	July 2023 - May 2026
	• Successfully completed B.Math(Hons) with distinction (76.67%) • Coursework: Real Analysis, Linear Algebra, Probability Theory, Elementary Number Theory, Numerical Computing, Statistical Inference and Hypothesis Testing, Introduction to Linear Models and Regression, Discrete Mathematics, Group Theory, Ring Theory, Ordinary Differential Equations, Analysis of Several Variables, Introduction to Stochastic Processes, Design and Analysis of Algorithms, Function Spaces, Information Theory, Complex Analysis, Optimization, Geometry.	
WBCHSE	Ramakrishna Mission Vivekananda Vidyamandir, Malda	2021 - 2023
	• Passed WBCHSE with 97%.	
WBBSE	Ramakrishna Mission Vivekananda Vidyamandir, Malda	2015 - 2021
	• Passed WBBSE with 96%.	

Achievements

• Successfully Completed three months internship at <i>Research Commons</i>	2025
• Selected for INSPIRE - SHE Fellowship	2023
• Qualified IAT with AIR 469.	2023
• Qualified National Science Examination for Chemistry.	2022
• Quallified IOQM 2020.	2020

Experience

- Completed CS50 Introduction to Cybersecurity (Harvard University)
- I mentored students for the upcoming ISI Entrance Examination at Ramakrishna Mission Vivekananda Vidyamandir, Malda during my winter vacation.
- Attended and successfully completed an online course of Advanced Theory in Statistics under Professor Xin Zhao. This course is organised by Beijing Institute of Mathematical Sciences and Application.
- Attended and successfully completed a course of Econometric Modelling and Business Analytics under Professor Jhareswar Maiti and Dr. Sayak Roychowdhury. The course included basics of Hypothesis Testing, Regression Analysis, ANOVA, ANCOVA, analysis of Auto Colinearity, Multicollinearity, Heteroscedasticity etc.
- Completed CS50x (Harvard University)

Skills

Languages	Bengali, English, Hindi, Sanskrit, German
Coding	C, Python, R, $\text{\LaTeX}$, OCTAVE/MATLAB
Other Activities	I am a member of the Math Club at ISI Bangalore. The goal of this club is to organize various talks, seminars and lecture series across the year. Besides, I am also the secretary of The Cultural Committee of ISI Bangalore. I was also the <i>Event Organizer and Manager</i> and <i>Member of the tech team</i> of CHAOS 2k25 , the annual fest of ISI Bangalore

Talks Given

- **Trapped or Free: Understanding Random Walk Dynamics** 2024
Math Club, ISI Bangalore
- **Probability Theory: Setting up the basic necessities** 2025
LIMIT Online Camp, ISI Bangalore
- **Random Fractal Percolation and Related Models** 2026
Math Club, ISI Bangalore

Conference Attended

International Conference for Supply Chain Modeling, Optimization, Digitalization & Analytics (ICoSMODA 2026)	2026
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Projects

1. I have done a research internship under Dr. Somnath Pradhan at IISER Bhopal on *Convex Optimization and its Connection to Markov Decision Processes*.
2. I completed a reading project on *Random Fractal Percolation, Russo-Seymour-Welsh Theory, Gaussian Fields* under Professor Rishideep Roy (University of Essex).
3. I have done a research internship under Dr. Amit Ghoshal at IISER Kolkata on *Analyzing transport properties of aquifers using percolation*.
4. I was one of the mentors of Undergraduate Group Reading Project: Winter 2024, organized by Math club of ISI Bangalore. The title of the Project was "Introduction to Dynamical Systems"

Publications and Preprints

1. S. Sarkar, A. Majumdar, M. Chatterjee, *A New Three-Players Auction Bridge with Dynamic Opponents and Team Members*, <https://arxiv.org/pdf/2508.05582>
2. S. Sarkar, A. Majumdar, M. Chatterjee, *3 - Players Auction Bridge, Statistical Algorithmic Strategies*, <https://arxiv.org/abs/2608.03217>
3. A. Majumdar, S. Sarkar, M. Chatterjee, *A Statistical Analysis of Three Player Auction Bridge*, <https://doi.org/10.13140/RG.2.2.28534.77127>

References

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